

Brac University

Department of Electrical & Electronic Engineering

Semester Spring-25

Course Number: EEE103L

Course Title: Computer Programming Laboratory

Section: 06

Lab Report



Experiment no. 03 (Part 2)

Name of the experiment: Familiarization with Decision Control Structures & Loop Control Structures.

Prepared by:

Name: **Tanzeel Ahmed** ID: **24321367**

Group Number: 04

Other Group members:

<i>Sl.</i>	<i>ID</i>	<i>Name</i>
1.	24121083	Abontika Das
2.	24121181	Tasnim Sultana
3.	24121249	Faria Rahman Oyshi
4.	24121251	Yeasin Khan Sifat

Lab report of experiment 3 (Part 2)

C. Write a program for a matchstick game being played between the computer and a user. Your program should ensure that the computer always wins. Rules for the game are as follows:

- ☐ There are 21 matchsticks.
- ☐ The computer asks the player to pick 1, 2, 3, or 4 matchsticks.
- ☐ After the person picks, the computer does its picking.
- ☐ Whoever is forced to pick up the last matchstick loses the game.

Code:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int stick = 21, player, computer;
```

```
    printf("Starting the matchstick game\n");
```

```
    printf("Rules: Pick 1, 2, 3, or 4 matchsticks on each turn. Whoever picks the last  
matchstick will lose.\n\n");
```

```
    while (stick > 1) {
```

```
        printf("Matchsticks left: %d\n", stick);
```

```
        printf("Pick 1, 2, 3, or 4 : ");
```

```
        scanf("%d", &player);
```

```
        if (player < 1 || player > 4)
```

```
        {
```

```
            printf("Wrong count. Pick between 1 to 4.\n");
```

```
        }
```

```
        stick = stick - player;
```

```
        if (stick == 1)
```

```
        {
```

```
            printf("Last Matchstick picked. You lose :D \n");
```

```
        }
```

```
        computer = 5 - player;
```

```
        printf("Computer picks %d matchstick.\n", computer);
```

```
        stick = stick - computer;
```

```

        if (stick == 1)
        {
            printf("Forced to pick the last one. You lose :D\n");
        }
    }
    return 0;
}

```

Screenshot of code:

The screenshot shows a C code editor with a file named 'matchstick.c'. The code implements a matchstick game where two players, 'player' and 'computer', take turns picking matchsticks from a pile of 21. The player who picks the last matchstick loses. The game logic is as follows:

- Initialization:** The game starts with 21 matchsticks. The player and computer are initialized.
- Game Loop:** A while loop runs as long as there are more than 1 matchstick left.
 - Player's Turn:** The player is prompted to pick 1, 2, 3, or 4 matchsticks. If the input is invalid (less than 1 or more than 4), the player is prompted to pick between 1 and 4.
 - Computer's Turn:** The computer picks a number of matchsticks (5 minus the player's pick) and prints the pick.
 - Matchstick Count:** The total number of matchsticks is updated after each pick.
 - Loss Check:** After each player's pick, the code checks if the number of matchsticks is now 1. If so, the player who just picked loses.
- Termination:** The game ends when the matchstick count reaches 1, and the player who picked it loses. The program returns 0.

```

1  #include <stdio.h>
2  int main()
3  {
4      int stick = 21, player, computer;
5
6      printf("Starting the matchstick game\n");
7      printf("Rules: Pick 1, 2, 3, or 4 matchsticks on each turn. Whoever picks the last matchstick will lose.\n\n");
8
9      while (stick > 1) {
10         printf("Matchsticks left: %d\n", stick);
11         printf("Pick 1, 2, 3, or 4 : ");
12         scanf("%d", &player);
13
14         if (player < 1 || player > 4)
15         {
16             printf("Wrong count. Pick between 1 to 4.\n");
17         }
18
19         stick = stick - player;
20
21         if (stick == 1)
22         {
23             printf("Last Matchstick picked. You lose :D \n");
24         }
25
26         computer = 5 - player;
27         printf("Computer picks %d matchstick.\n", computer);
28
29         stick = stick - computer;
30
31         if (stick == 1)
32         {
33             printf("Forced to pick the last one. You lose :D\n");
34         }
35     }
36     return 0;
37 }

```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\extensions\ms-vscode.cpptools-1.23.6-win32-x64\debug
in=Microsoft-MIEngine-In-jvhvwznu.bzu' '--stdout=Microsoft-MIEngine-Out-aans1yyo.2vq' '--stderr=Mi
crosoft-MIEngine-Pid-0adzyd3z.rb3' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe' '--interpreter=mi'
Starting the matchstick game
Rules: Pick 1, 2, 3, or 4 matchsticks on each turn. Whoever picks the last matchstick will lose.

Matchsticks left: 21
Pick 1, 2, 3, or 4 : 3
Computer picks 2 matchstick.
Matchsticks left: 16
Pick 1, 2, 3, or 4 : 2
Computer picks 3 matchstick.
Matchsticks left: 11
Pick 1, 2, 3, or 4 : 3
Computer picks 2 matchstick.
Matchsticks left: 6
Pick 1, 2, 3, or 4 : 4
Computer picks 1 matchstick.
Forced to pick the last one. You lose :D
```

d. Write a program to produce the following output:

```

  1
 2 3
4 5 6
 7 8 9 10
```

Code:

```
#include<stdio.h>
```

```
int main()
{
    printf("    1\n");
    printf("   2 3\n");
    printf("  4 5 6\n");
    printf("   7 8 9 10\n");

    return 0;
}
```

Screenshot of code:

```
D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > I
1  #include<stdio.h>
2
3  int main()
4  {
5      printf("        1\n");
6      printf("        2 3\n");
7      printf("        4 5 6\n");
8      printf("        7 8 9 10\n");
9
10     return 0;
11
12 }
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\bin\Microsoft-MIEngine-In-drlfgrpj4c' '--s
Microsoft-MIEngine-Pid-a3dkvu4x.qnm' '--dbg

        1
        2 3
       4 5 6
      7 8 9 10
```

e. Population of a town today is 100000. The population has increased steadily at the rate of 10 % per year for last 10 years. Write a program to determine the population at the end of each year in the last decade.

Code:

```
#include <stdio.h>
```

```
int main()
```

```

{
    double p, g;
    int i;

    p=100000;
    g= 0.10;

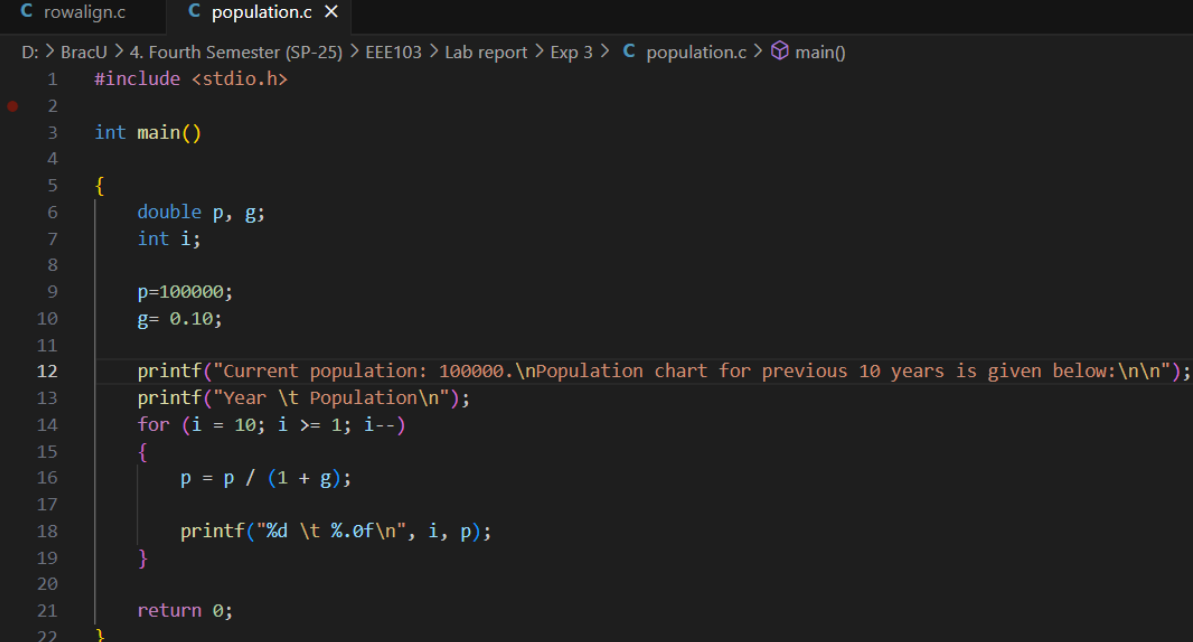
    printf("Current population: 100000.\nPopulation chart for previous 10 years is given
below:\n\n");
    printf("Year \t Population\n");
    for (i = 10; i >= 1; i--)
    {
        p = p / (1 + g);

        printf("%d \t %.0f\n", i, p);
    }

    return 0;
}

```

Screenshot of code:



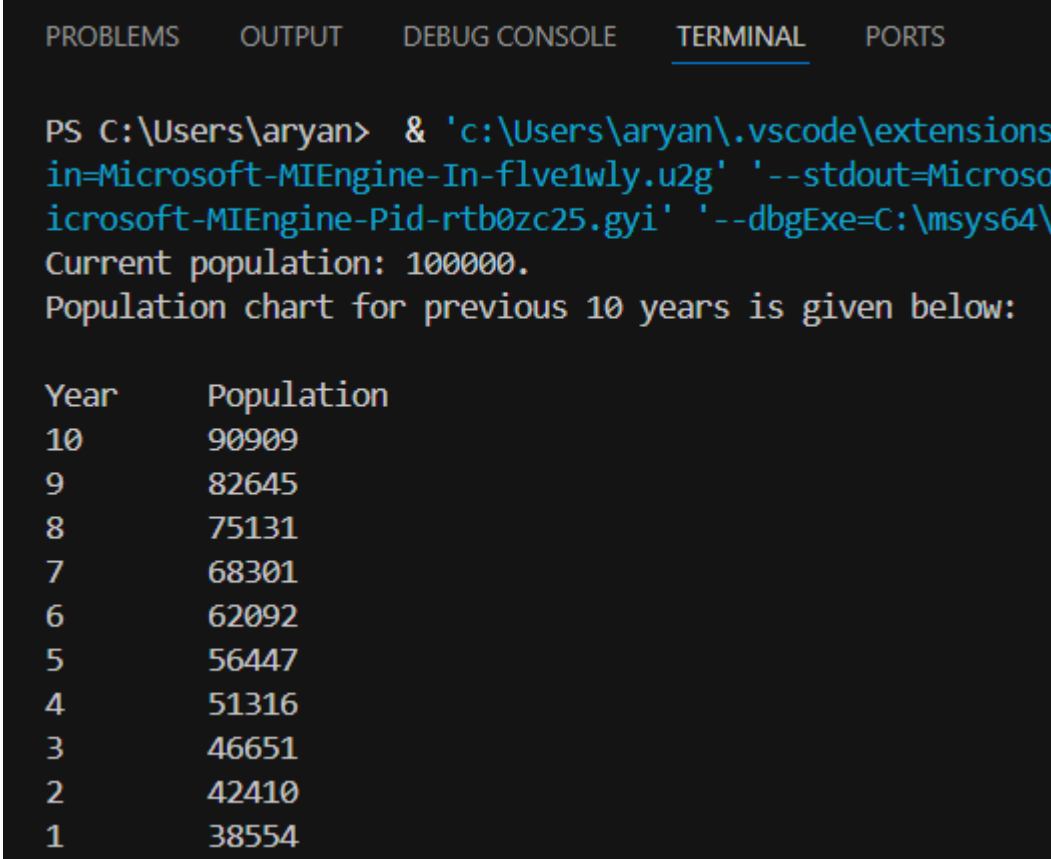
The screenshot shows a code editor with two tabs: 'rowalign.c' and 'population.c'. The 'population.c' tab is active, displaying the following C code:

```

D: > BracU > 4. Fourth Semester (SP-25) > EEE103 > Lab report > Exp 3 > population.c > main()
1  #include <stdio.h>
2
3  int main()
4  {
5      double p, g;
6      int i;
7
8      p=100000;
9      g= 0.10;
10
11     printf("Current population: 100000.\nPopulation chart for previous 10 years is given below:\n\n");
12     printf("Year \t Population\n");
13     for (i = 10; i >= 1; i--)
14     {
15         p = p / (1 + g);
16
17         printf("%d \t %.0f\n", i, p);
18     }
19
20     return 0;
21 }
22

```

Output:



The screenshot shows a VS Code terminal window with the 'TERMINAL' tab selected. The terminal displays the output of a command executed in a PowerShell prompt. The command is a complex call to a program, likely a simulation, with various flags for output and debugging. The output shows the current population and a table of population data for the previous 10 years.

```
PS C:\Users\aryan> & 'c:\Users\aryan\.vscode\extensions  
in=Microsoft-MIEngine-In-flve1wly.u2g' '--stdout=Microso  
icrosoft-MIEngine-Pid-rtb0zc25.gyi' '--dbgExe=C:\msys64\  
Current population: 100000.  
Population chart for previous 10 years is given below:
```

Year	Population
10	90909
9	82645
8	75131
7	68301
6	62092
5	56447
4	51316
3	46651
2	42410
1	38554